

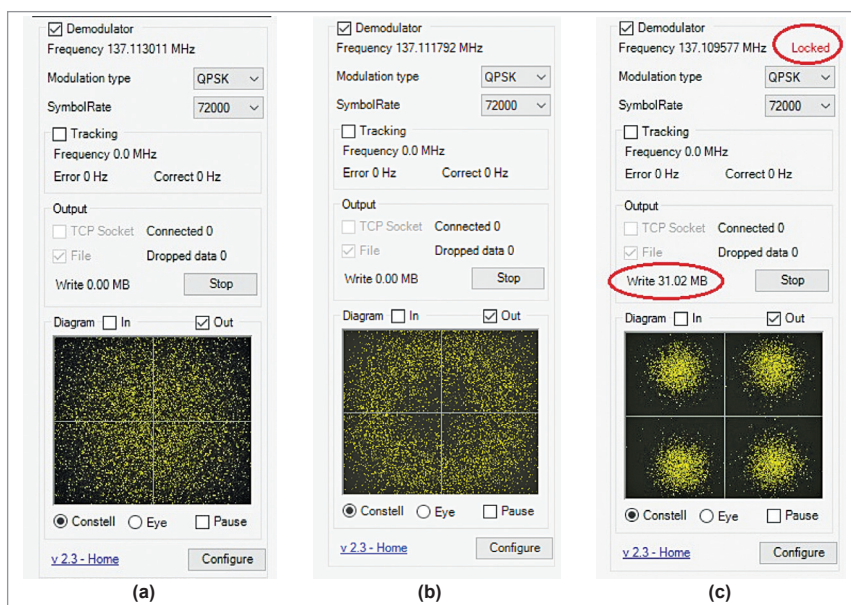


Fig. 5: (a) No signal, (b) Weak signal, and (c) Carrier locked signal

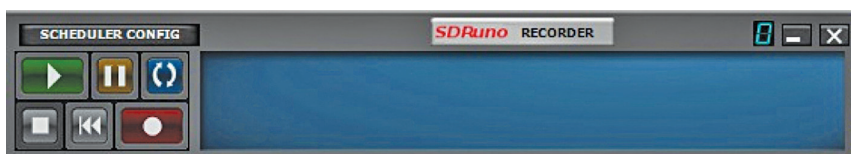


Fig. 6(a): SDRUno recorder

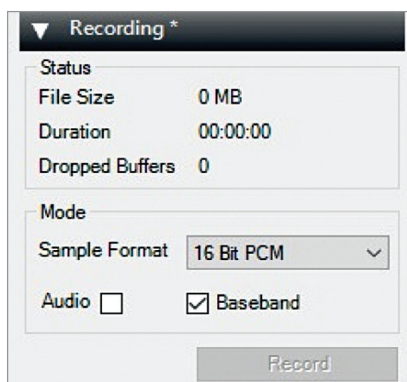


Fig. 6(b): RTL-SDR recorder with SDR#

demodulated data. Fig. 2 illustrates a flow-chart for both these options.

The RTL-SDR with SDR# software and Meteor plug-in. The readers should familiarise themselves with the installation and usage of the SDR# software from AirSpy for use with the RTL-SDR receiver and, more importantly, the Meteor plug-in. One can download the SDR# software and add the plug-in later or download the latest version of

the software from www.airspy.com. The Meteor M2 demodulator plug-in can be found at http://happysat.nl/meteor_2.3.zip

The author is grateful to Alex of the HappySat.nl website and Les Hamilton of the UK. Their articles are very useful for setting up the Meteor satellite demodulation and processing software properly. The off-line Meteor LRPT analyser can also be found at <https://leshamilton.co.uk/soft/Meteor-LRPT-Suite.1.8.1.zip>

Given below are the steps to set up the software to receive the satellite signals:

Step 1: Receiver preparation and settings. The Meteor demodulator is a QPSK soft demodulator. Also, the software defined radio is nothing but an ADC sampling the RF signal at some high rates of > 1MSPS (mega samples per second).

Therefore, we need some settings at the SDR sampling rate as well for

the demodulator to work properly. After studying various published information, we had set the RTL-SDR sampling rate at 1.4MSPS, somewhere in between 1MSPS and 2MSPS. This can be achieved by clicking the Configure Source button on the SDR# main menu at the top. A dialogue box appears, as shown in Fig. 3(a). Maintain settings accordingly with the RTL AGC box unticked and Tuner AGC RF Gain set to the maximum. The oscillator frequency correction needs to be set around 70ppm, or the value specified by the manufacturer, for better frequency stability.

Next, set the radio receiver in wide-band FM (WFM) with a bandwidth of around 225kHz and tick the Correct I/Q check box, as shown in Fig. 3(b).

Step 2: The Meteor demodulator settings. Setup the demodulator plug-in. If your demodulator plug-in is successfully installed, it appears in your list as shown in Fig. 4(a). When you click on it, you should see the demodulator settings, as shown in Fig. 4(b).

As shown in Fig. 4(b), set the modulation as QPSK and the symbol rate to either 72k or auto. Uncheck the Demodulator and Tracking boxes at this stage. The tracking option is utilised when the Doppler shifts in frequency or in conjunction with an automatic antenna tracker. The demodulator check-box enables or disables the demodulator and can be used at the time of receiving the satellite.

For output settings, check the File check-box, so that the demodulator writes the demodulated data as a file. The folder or location where this file will be stored is set using the Configure settings at the bottom-right of the diagram. When the demodulator starts writing data to the file, it will show the data size being written to the file. For a successful satellite file this data can range from 36MB to all the way up